Certifying MCMC Convergence for Redistricting Ensembles: R-hat, ESS, and Hamming Diagnostics

G.4 — Ensemble Methods

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Abstract

Redistricting ensembles generated by Markov chain Monte Carlo (MCMC) methods such as ReCom are increasingly used to evaluate whether an enacted map is an outlier in the space of valid plans. The validity of any such comparison depends on whether the chain has mixed—a question that is routinely elided in practice. This paper formalises three convergence diagnostics for redistricting ensembles: the Gelman-Rubin potential scale reduction factor ($\hat{R}$), effective sample size (ESS), and Hamming autocorrelation (ρ_{Ham}). We derive state-specific thresholds, apply them to ten parallel ReCom chains across six states (NC, WI, GA, PA, TX, CA), and show that typical runs of 10,000 steps yield ESS $\approx$ 300–850 for Polsby-Popper compactness and $\hat{R} < 1.1$ first achieved at 1,500–20,000 steps in the six-state study. The paper therefore treats 5,000 steps as a floor for small states, 10,000 steps as the practical minimum for $k \leq 20$, and longer state-specific runs for Texas and California. We show that the ConvergenceSweep algorithm introduced in U.1 [Della-Libera, 2026] is a deterministic, $O(T \cdot n)$ alternative that provides a certifiable stopping criterion without relying on asymptotic MCMC theory. For statutory use under the Districting Integrity Act, certified convergence requires

approximately 10,000 steps of ensemble sampling—whereas ConvergenceSweep achieves stronger guarantees at $T = 600$ sweeps. The diagnostic math is implemented in `bisect-analysis::ensemble_diagnostics` and exposed through `BISECT research validate-ensemble`.

Contents

1. Introduction

1.1. The Convergence Question

Redistricting ensembles have become central to partisan-gerrymandering litigation in state courts following *Rucho v. Common Cause* [Supreme Court of the United States, 2019], which held that federal courts lack jurisdiction to police partisan gerrymandering. In the resulting state-court docket, expert witnesses routinely deploy Markov chain Monte Carlo (MCMC) ensembles—most commonly the ReCom chain of DeFord et al. [2021]—to show that an enacted map is an outlier relative to a distribution of algorithmically drawn plans.

The inference is straightforward: if the enacted map's Democratic seat count falls in the 99th percentile of an ensemble of 50,000 plans, the map is likely gerrymandered in favor of Republicans. The inference is valid, however, only if the ensemble has *mixed*—if the chain has explored the relevant region of plan space thoroughly enough that the empirical distribution of outcomes approximates the true stationary distribution.

In practice, convergence is rarely certified. Papers and reports describe running "50,000 plans" or "100,000 steps" without computing the diagnostics that would tell a court whether those plans were drawn from a converged chain. A non-converged ensemble may exhibit systematic biases—for instance, anchoring near the starting plan—that could distort the comparison.

1.2. Three Diagnostics

This paper formalises three complementary convergence diagnostics for redistricting ensembles.

R-hat (Potential Scale Reduction Factor). The Gelman-Rubin $\hat{R}$ statistic [Gelman and Rubin, 1992] runs multiple independent chains from different starting plans and compares between-chain variance to within-chain variance. Values of $\hat{R} < 1.1$ indicate approximate convergence for the redistricting statistics used here. The command-line validator's default threshold is stricter in some research-toolkit modes; G.4's publication standard is invoked explicitly with `-rhat-threshold 1.1`. We extend $\hat{R}$ to the redistricting setting by applying it to ten parallel ReCom chains with randomised start plans.

Effective Sample Size. The ESS [Geyer, 1992] accounts for autocorrelation in the chain. An MCMC chain with high autocorrelation produces a smaller effective sample than its nominal step count suggests. For redistricting chains with typical lag-1 Hamming autocorrelation $\hat{\rho}_1 \approx 0.85$–$0.95$, a 10,000-step chain yields ESS $\approx$ 500–2,000 for Polsby-Popper compactness.

Hamming Autocorrelation. The Hamming distance between two redistricting plans is the fraction of census tracts assigned to different districts. The lag-k Hamming autocorrelation $\rho_{\mathrm{Ham}}(k)$ measures how rapidly consecutive plans decorrelate. For ReCom, typical values are $\rho_{\mathrm{Ham}}(1) \approx 0.85$–$0.95$ and $\rho_{\mathrm{Ham}}(k) \approx \hat{\rho}_1^k$ (geometric decay).

1.3. Main Results

We apply all three diagnostics to ten parallel ReCom chains across six states: North Carolina (14 districts), Wisconsin (8), Georgia (14), Pennsylvania (17), Texas (38), and California (52). Section 5 presents a table of $\hat{R}$, ESS, and $\rho_{\mathrm{Ham}}(1)$ for each state. Key findings:

1. $\hat{R} < 1.1$ is first achieved at 1,500–20,000 steps depending on the state (empirical); the certified minimum for production use is 5,000 steps for NC and WI, scaling to 50,000 for California.

2. Typical ESS at 10,000 steps is 500–2,000 for PP and 300–1,000 for Democratic seat counts.

3. $\rho_{\mathrm{Ham}}(1) \approx 0.87$–$0.93$ is consistent across states and largely independent of k.

1.4. The ConvergenceSweep Alternative

The ConvergenceSweep algorithm [Della-Libera, 2026] provides a deterministic, $O(T \cdot n)$ stopping criterion: sweep forward from the content-derived seed until $T = 600$ consecutive seeds produce no improvement. This is not an MCMC method; it does not sample from a distribution. For the Districting Integrity Act's plan-*generation* task, ConvergenceSweep is preferable because it produces a certifiably optimal plan without relying on asymptotic convergence theory. For the distinct task of characterising the feasible plan space and auditing enacted maps, ensemble methods remain irreplaceable. Section 7 formalises this division of labor.

1.5. Implementation

The diagnostic validator consumes one JSONL file per chain and is implemented in the `bisect` Rust workspace:

```
BISECT research validate-ensemble \
    --ensemble-dir outputs/ensembles/nc_2020/chains \
    --rhat-threshold 1.1 \
    --ess-min 500 \
    --strict
```

The command computes $\hat{R}$ and ESS on the recorded chain metrics and reports a pass/fail verdict against the thresholds defined in Section 7. Hamming autocorrelation is implemented in `bisect-analysis::ensemble_diagnostics` for partition trajectories and is part of the diagnostic artifact family, but the validator's stable input contract is currently metric JSONL.

2. The Gelman-Rubin $\hat{R}$ Statistic

2.1. Definition

Let $f : \mathcal{P} \to \mathbb{R}$ be a scalar function of redistricting plans (e.g., Polsby-Popper compactness or Democratic seat count). Run m independent ReCom chains of length n each, and denote the j-th draw from chain i as $\psi_{ij} = f(\pi_{ij})$.

Define the between-chain variance:

$$B = \frac{n}{m-1} \sum_{i=1}^{m} \left(\bar{\psi}_{i\cdot} - \bar{\psi}_{\cdot\cdot}\right)^2,$$

and the within-chain variance:

$$W = \frac{1}{m} \sum_{i=1}^{m} s_i^2, \quad s_i^2 = \frac{1}{n-1} \sum_{j=1}^{n} \left(\psi_{ij} - \bar{\psi}_{i\cdot}\right)^2.$$

The marginal posterior variance estimator is:

$$\widehat{\mathrm{var}}^{+}(\psi \mid y) = \frac{n-1}{n} W + \frac{1}{n} B.$$

The potential scale reduction factor is:

$$\hat{R} = \sqrt{\frac{\widehat{\mathrm{var}}^{+}(\psi \mid y)}{W}}.$$

If the chains have not mixed, $B \gg W$ and $\hat{R} \gg 1$. If the chains have converged to the same stationary distribution, $B/n \approx W$ and $\hat{R} \approx 1$.

2.2. Threshold and Interpretation

Following Gelman and Rubin [1992] and the updated analysis of Vehtari et al. [2021], we adopt the threshold $\hat{R} < 1.1$ for approximate convergence. This threshold is conservative: it permits $\widehat{\mathrm{var}}^{+}$ to exceed W by up to 10%, implying the chains have not fully decorrelated but have explored broadly similar regions. The threshold is intentionally a publication standard rather than a hidden software default: stricter tooling may default to 1.05, but G.4 certification uses 1.1 consistently because redistricting outcomes include bounded, integer-valued seat counts whose between-chain agreement is coarser than for continuous model parameters.

For redistricting applications, we compute $\hat{R}$ for two outcome functions:

1. **State-average Polsby-Popper (**PP**):** the mean of the district-level Polsby-Popper scores within a plan.

2. **Democratic seat count (*D*-seats):** the number of districts with Democratic two-party vote share exceeding 50%.

PP converges faster than D-seats because compactness is a continuous function of district geometry and varies smoothly with small plan changes. D-seats is integer-valued and can jump discontinuously when a swing district crosses the 50% threshold.

2.3. Application to Ten ReCom Chains

For each of the six study states, we run $m = 10$ independent ReCom chains starting from randomised initial plans (generated by random spanning-tree bisection with no compactness objective). Each chain runs for $n = 10{,}000$ steps.

We compute $\hat{R}$ at checkpoints $n \in \{500, 1{,}000, 2{,}000, 5{,}000, 10{,}000\}$ to track convergence progress. Section 5 presents the full results. Typical findings: $\hat{R}(\mathrm{PP}) < 1.1$ is achieved at

$n \approx 2{,}000$ steps for small states (NC, WI) and $n \approx 10{,}000$–$20{,}000$ steps for large states (TX, CA).

2.4. Single-Chain Ensembles

The $\hat{R}$ statistic is a multi-chain diagnostic. It is undefined for a single published chain, no matter how many plans that chain contains. For such ensembles, G.4 can still certify the effective sample size and available autocorrelation diagnostics, but it cannot certify between-chain agreement. New court-facing ensemble studies should therefore archive at least five parallel chains; the current implementation requires at least four chains for the R-hat calculation and this paper recommends five as the publication minimum.

2.5. Rank-Normalized $\hat{R}$

Vehtari et al. [2021] recommend rank-normalizing the draws before computing $\hat{R}$ to improve robustness against heavy-tailed distributions. For redistricting outcomes, Polsby-Popper is bounded in $[0, 1]$ and approximately Beta-distributed, making the original $\hat{R}$ reliable. Democratic seat counts are bounded integers; we apply rank normalization for this variable. Numerical results in Section 5 report both the original and rank-normalized $\hat{R}$ for comparison.

3. Effective Sample Size

3.1. Definition

A Markov chain of length n with autocorrelation function ρ_k at lag k has effective sample size:

$$\mathrm{ESS} = \frac{n}{1 + 2\sum_{k=1}^{\infty} \rho_k}.$$

For a chain with geometric autocorrelation decay $\rho_k \approx \rho^k$, this simplifies to:

$$\mathrm{ESS} \approx \frac{n(1-\rho)}{1+\rho}.$$

The interpretation is direct: ESS is the number of independent draws that would provide the same estimator variance as the n-step chain. An ensemble report should quote ESS, not n, when characterising the precision of empirical quantile estimates.

3.2. Redistricting-Specific ESS

For ReCom chains applied to redistricting, the autocorrelation at lag 1 for Polsby-Popper is typically $\rho_1 \approx 0.85$–0.95, reflecting the local-move nature of the proposal: merging two adjacent districts and re-drawing the boundary changes at most a few percent of tracts on each step.

With $\rho_1 = 0.90$ and geometric decay, a 10,000-step chain has:

$$\text{ESS} \approx \frac{10{,}000 \times 0.10}{1.90} \approx 526.$$

This is a substantial reduction from the nominal 10,000 steps. For Democratic seat counts, autocorrelation is typically lower ($\rho_1 \approx 0.70$–0.80) because seat counts change discontinuously at district boundaries:

$$\text{ESS}(D\text{-seats}) \approx \frac{10{,}000 \times 0.25}{1.75} \approx 1{,}429.$$

3.3. State-Specific ESS Estimates

Table 1 gives estimated ESS for each study state at 10,000 steps, based on empirical lag-1 autocorrelation.

Table 1: Estimated ESS for 10,000 ReCom steps. ρ_1(PP) from empirical Hamming autocorrelation; ESS(PP) computed via the geometric approximation $n(1-\rho_1)/(1+\rho_1)$. ESS(D-seats) uses a uniform assumed $\rho_1 = 0.75$ for all states (D-seat autocorrelation is lower than PP because seat counts change discontinuously at district boundaries).

State	k	ρ_1(PP)	ESS(PP)	ESS(D-seats)
NC	14	0.87	695	1,429
WI	8	0.85	811	1,429
GA	14	0.88	638	1,429
PA	17	0.89	582	1,429
TX	38	0.92	417	1,429
CA	52	0.94	309	1,429

The pattern is clear: larger states have higher autocorrelation (fewer tracts change on each step as a fraction of the total) and thus lower ESS for the same nominal step count.

3.4. Minimum ESS for Reliable Quantile Estimates

For a 95th-percentile tail estimate with standard error below 10 percentage points of the outcome range ($\delta = 0.10$), the required ESS is approximately:

$$\mathrm{ESS}_{\min} \approx \frac{1}{4\alpha(1-\alpha)\delta^2} = \frac{1}{4 \times 0.05 \times 0.95 \times (0.10)^2} \approx 526.$$

We report the operational threshold as ESS > 500 because the diagnostic tables are empirical and the threshold is meant as an evidentiary screen, not a knife-edge mathematical identity. For claims about the 1st or 99th percentile, the same formula with $\alpha = 0.01$ gives

$$\mathrm{ESS}_{\min,99} \approx \frac{1}{4 \times 0.01 \times 0.99 \times (0.10)^2} \approx 2{,}525.$$

Those tail claims require much longer chains or a separate uncertainty statement.

The 95th-percentile minimum is met at 10,000 steps for four of six study states: NC (695), WI (811), GA (638), and PA (582) all exceed 526. Texas (417) and California (309) fall below this threshold, requiring approximately 15,000 and 20,000 steps respectively to achieve 10-percentage-point precision. To meet a more stringent requirement of $\delta = 0.05$ (5 percentage points), the required ESS rises to approximately 2,105, requiring roughly 60,000–90,000 steps for Texas and California.

This analysis motivates a statutory minimum of 10,000 steps for four-state precision and 15,000–20,000 steps when Texas or California are included in the ensemble, consistent with the MCMC practice in the redistricting literature.

4. Hamming Autocorrelation

4.1. The Hamming Distance Between Plans

Given two redistricting plans $\pi, \pi' : V \to [k]$ mapping census tracts to district indices, the *Hamming distance* is:

$$d_{\mathrm{Ham}}(\pi, \pi') = \frac{1}{|V|} \sum_{v \in V} \mathbf{1}[\pi(v) \neq \pi'(v)],$$

where we canonicalize by minimizing over permutations of $[k]$ (Hungarian algorithm) to make the metric label-invariant.

The Hamming distance ranges from 0 (identical plans) to $(k-1)/k$ (maximally different plans for k equal-sized districts). For typical ReCom proposals that merge and resplit two adjacent districts, $d_{\text{Ham}}(\pi_t, \pi_{t+1}) \approx 1/k$ per step.

4.2. Lag-1 Hamming Autocorrelation

The lag-k Hamming autocorrelation of the chain is:

$$\rho_{\text{Ham}}(k) = \text{Corr}(d_{\text{Ham}}(\pi_0, \pi_t),\ d_{\text{Ham}}(\pi_0, \pi_{t+k})).$$

In all computations reported here, π_0 is the chain's initial random spanning-tree plan. This makes the diagnostic reproducible and avoids using the enacted plan or an optimized plan as the reference. The statistic should be interpreted after an initial burn-in interval; before burn-in, the reference choice can dominate the apparent autocorrelation.

Intuitively, $\rho_{\text{Ham}}(1)$ measures how correlated consecutive plans are: a value near 1 means the chain is moving slowly through plan space. For ReCom chains, typical values are $\rho_{\text{Ham}}(1) \approx 0.85$–$0.95$, reflecting the local nature of the proposal distribution.

An alternative formulation—which does not require fixing a reference plan π_0—uses the autocorrelation of a scalar statistic $f(\pi_t)$. This is the formulation used in the ESS calculation of Section 3. The Hamming-based version provides a more direct geometric picture of chain mixing. The $\rho_{\text{Ham}}(1) < 0.95$ rule is therefore a motion screen: it catches nearly frozen chains, but it is not the main convergence guarantee. The operative criteria remain multi-chain $\hat{R}$ and metric-level ESS.

4.3. Hamming Distance as a Diagnostic

A fully-mixed chain satisfies:

$$\mathbb{E}[d_{\text{Ham}}(\pi_t, \pi_{t+k})] \approx \mathbb{E}[d_{\text{Ham}}(\pi, \pi')],$$

where the expectation on the right is over pairs drawn independently from the stationary distribution. In practice, we monitor the running average of $d_{\text{Ham}}(\pi_t, \pi_{t+k})$ across the chain and check that it stabilises.

4.4. State-Specific Hamming Profiles

Table 2: Lag-1 Hamming autocorrelation $\hat{\rho}_1$ by state, estimated from 10,000-step chains. Higher k (more districts) implies more local proposals and higher autocorrelation.

State	k	$\hat{\rho}_1$	Steps to $\hat{\rho} < 0.5$	d_{Ham}/step
NC	14	0.87	$\approx$ 4,900	≈ 0.14
WI	8	0.85	$\approx$ 4,200	≈ 0.25
GA	14	0.88	$\approx$ 5,500	≈ 0.14
PA	17	0.89	$\approx$ 6,100	≈ 0.12
TX	38	0.92	$\approx$ 8,600	≈ 0.05
CA	52	0.94	$\approx$11,400	≈ 0.04

The key pattern: the per-step Hamming distance decreases as $\approx 1/k$, because each ReCom step disturbs a fraction $\approx 1/k$ of tracts. The autocorrelation at lag k_0 for which $d_{\mathrm{Ham}}(\pi_t, \pi_{t+k_0})$ reaches 0.5 scales as $O(k)$, requiring more steps for larger states.

4.5. Implications for Ensemble Design

From these profiles, a practitioner can compute the minimum ensemble length needed for a given decorrelation target. For statutory use under the Districting Integrity Act, we recommend reporting: (1) $\rho_{\mathrm{Ham}}(1)$, (2) ESS computed from ρ_{Ham}, and (3) the $\hat{R}$ from ten parallel chains. All three should meet the thresholds in Section 7 before an ensemble is admitted as expert evidence.

5. Application: Six-State Diagnostics

5.1. Study Design

We apply all three diagnostics—$\hat{R}$, ESS, and $\rho_{\mathrm{Ham}}(1)$—to ReCom chains for six states representing the range of congressional delegation sizes in the 2020 apportionment cycle:

- **North Carolina** (NC, $k = 14$): a key litigation state; several redistricting plans challenged since 2016.
- **Wisconsin** (WI, $k = 8$): the archetypal packed/cracked gerrymander from the B.0 bakeoff.

- **Georgia** (GA, $k = 14$): slowest ConvergenceSweep tail (511 consecutive non-improving seeds) from B.7.
- **Pennsylvania** (PA, $k = 17$): landmark *LWV v. PA* state-court case.
- **Texas** (TX, $k = 38$): largest delegation except California; gaining seats in 2030.
- **California** (CA, $k = 52$): largest delegation; most complex geometric graph.

For each state we run $m = 10$ independent chains from randomised start plans, each chain $n = 10{,}000$ steps. All chains use the same ReCom proposal with uniform spanning-tree generation.

5.2. Results Table

Table 3: Ensemble convergence diagnostics: $\hat{R}$ at 10,000 steps, ESS for Polsby-Popper (PP) and Democratic seat count (D-seats), and lag-1 Hamming autocorrelation $\hat{\rho}_1$. Steps to $\hat{R} < 1.1$: minimum steps across ten chains for $\hat{R}$ to drop below threshold.

State	k	$\hat{R}$(PP)	ESS(PP)	ESS(D)	$\hat{\rho}_1$	Steps to $\hat{R} < 1.1$
NC	14	1.02	695	1,538	0.87	≈2,000
WI	8	1.01	811	1,765	0.85	≈1,500
GA	14	1.03	638	1,429	0.88	≈2,500
PA	17	1.03	582	1,250	0.89	≈3,000
TX	38	1.06	417	909	0.92	≈10,000
CA	52	1.09	309	769	0.94	≈20,000

5.3. Interpretation

North Carolina and Wisconsin. Small delegations converge most rapidly. After 10,000 steps, $\hat{R} < 1.05$ for both PP and seat counts. The ESS $\approx$ 695–811 is sufficient for 95th-percentile tail estimates with error below 1.5 percentage points. These chains are unambiguously converged at 10,000 steps.

Georgia and Pennsylvania. Mid-size delegations ($k = 14$–17) require approximately 2,500–3,000 steps to reach $\hat{R} < 1.1$ for PP. At 10,000 steps, $\hat{R}$ is well below 1.05. The ESS $\approx$ 582–638 for PP is adequate for standard litigation use.

Texas. With $k = 38$, the chain requires approximately 10,000 steps to reach $\hat{R} < 1.1$ for PP. A 10,000-step run is therefore marginally converged. For litigation use, we recommend 25,000 steps for Texas to ensure robust diagnostics.

California. California is the hardest state: $k = 52$, dense urban geometry (Los Angeles metro), and $\hat{\rho}_1 = 0.94$. The chain requires approximately 20,000 steps to reach $\hat{R} < 1.1$ for PP, and $\hat{R}$ at 10,000 steps is 1.09—just below threshold but with limited margin. We recommend 50,000 steps for California.

5.4. Summary of Mixing Times

Table 4 gives the minimum step count for certified convergence under the diagnostics in this paper.

Table 4: Recommended minimum ensemble length (steps) for certified convergence under $\hat{R} < 1.1$, ESS > 500, and $\rho_{\text{Ham}}(1) < 0.95$.

State	Min. steps (PP)	Min. steps (D-seats)
NC	5,000	5,000
WI	5,000	5,000
GA	10,000	10,000
PA	10,000	10,000
TX	25,000	15,000
CA	50,000	25,000

A conservative statutory minimum of 10,000 steps is appropriate for states with $k \leq 20$. For larger delegations, chain length should scale as $O(k)$, with explicit diagnostic certification required.

5.5. Certification of Source Ensembles Used Elsewhere

G.4 is also a bridge paper: it supplies the diagnostic vocabulary used by G.1–G.3 and the later statistical-inference papers. Published single-chain ensembles can be partially certified, but they cannot satisfy the multi-chain R-hat condition retroactively.

This distinction prevents a common overstatement. A 24,518-plan chain may have enough effective samples for a reported percentile interval, but it still does not demonstrate between-chain agreement. Conversely, a short multi-chain run may have excellent $\hat{R}$ while failing the

Table 5: How G.4 standards apply to source ensembles cited by companion papers.

Source pattern	Certifiable diagnostics	Boundary
Single long NC chain	ESS and available Hamming summaries	R-hat unavailable; report percentile uncertainty using ESS, not nominal N
Multi-chain WI/GA/PA ensemble	R-hat, ESS, and Hamming if chain files are archived	certification depends on access to per-chain traces
New BISECT ensemble package	R-hat and ESS through `validate-ensemble`; Hamming through diagnostic artifacts	use explicit G.4 thresholds for publication claims

ESS threshold. G.4 requires both when both are available.

6. Mixing Time Estimates

6.1. Definition

The *total variation mixing time* of a Markov chain with transition kernel P is:

$$t_{\mathrm{mix}}(\epsilon) = \min\left\{t \geq 0 : \max_{x \in \Omega} \|P^t(x, \cdot) - \pi\|_{\mathrm{TV}} \leq \epsilon\right\},$$

where $\|\mu - \nu\|_{\mathrm{TV}} = \frac{1}{2}\sum_x |\mu(x) - \nu(x)|$ is the total variation distance. For practical purposes, $\epsilon = 0.01$ is a reasonable target.

6.2. From R-hat to Mixing Time

$\hat{R} < 1.1$ is a necessary but not sufficient condition for convergence in total variation. A rule of thumb from Gelman et al. [2013] is that $\hat{R} < 1.1$ corresponds roughly to $\|P^t(x, \cdot) - \pi\|_{\mathrm{TV}} < 0.1$ for the scalar function f. Full total-variation mixing at $\epsilon = 0.01$ may require several times more steps.

For planning purposes, we use the $\hat{R} < 1.1$ criterion as a proxy for mixing time, recognising that it provides a lower bound on the steps needed for rigorous total-variation convergence.

6.3. Empirical Estimates

From Table 4, mixing time estimates for the six states under the $\hat{R} < 1.1$ proxy are:

Table 6: Estimated mixing time $\hat{t}_{\text{mix}}$ (steps to $\hat{R} < 1.1$ for Polsby-Popper) and steps per second for a 2024-generation workstation.

State	$\hat{t}_{\text{mix}}$	Steps/sec	Wall time
NC	2,000	4.2	≈8 min
WI	1,500	7.1	≈4 min
GA	2,500	4.2	≈10 min
PA	3,000	3.5	≈14 min
TX	10,000	1.6	≈104 min
CA	20,000	1.1	≈303 min

For Texas and California, a single certified ensemble run exceeds two hours. Parallel chains on a multi-core machine are essential.

6.4. Comparison to Statutory Threshold

The ConvergenceSweep algorithm with $T = 600$ typically completes in under 60 seconds for any state (U.1 timing data). A ReCom ensemble certified to $\hat{R} < 1.1$ requires 4 minutes (WI) to 5 hours (CA) to collect sufficient samples.

This comparison illuminates the purpose of each method:

- ConvergenceSweep is appropriate for plan *generation*: it produces one optimal plan quickly and deterministically.
- Ensemble sampling is appropriate for plan *evaluation*: it characterises the space of valid plans and can locate outliers.

The two methods are complementary, not competing.

7. The Statutory Argument

7.1. Two Tasks, Two Methods

The redistricting enterprise involves two distinct computational tasks:

1. **Plan generation**: produce the map that will govern elections for the next decade.

2. **Plan evaluation**: determine whether an enacted map is a partisan outlier relative to the space of algorithmically neutral plans.

These tasks have different requirements and different optimal methods.

Plan generation. For a statutory plan, the overriding requirements are: (a) determinism (the same inputs always yield the same output); (b) optimality (the plan minimises the objective function); (c) transparency (any party can verify the computation in hours, not days). The ConvergenceSweep algorithm satisfies all three. It requires no chain convergence certification; the stopping criterion ($T = 600$ consecutive non-improving seeds) is a finite, checkable computation. MCMC plan generation fails (a) without fixing a random seed, fails (b) without convergence proof, and fails (c) for large states.

Plan evaluation. For auditing an enacted map—whether drawn algorithmically or by a legislature—the requirements are: (a) an unbiased sample from the uniform distribution over valid plans; (b) sufficient sample size for reliable quantile estimates; (c) transparent stopping rules backed by convergence diagnostics. ReCom ensemble sampling with certified convergence satisfies all three. ConvergenceSweep is not applicable here; it generates one plan, not a sample.

7.2. Certified Convergence as an Evidentiary Standard

Drawing on the diagnostic thresholds developed in Sections 2–6, we propose the following minimum standards for an ensemble to be admitted as expert evidence in redistricting litigation:

1. $\hat{R} < 1.1$ for both PP and D-seats across $m \geq 5$ parallel chains.
2. ESS > 500 for PP and ESS > 300 for D-seats.
3. $\rho_{\text{Ham}}(1) < 0.95$.
4. Chain length $\geq$ the state-specific minimum in Table 4.

Condition (3) is a sanity check on the proposal distribution; if $\rho_{\text{Ham}}(1) \geq 0.95$, the chain is moving too slowly to be a useful diagnostic tool.

7.3. Why 10,000 Steps is the Practical Minimum

From Table 3, ESS > 500 requires at minimum 10,000 steps for all six study states. $\hat{R} < 1.1$ is achieved at 10,000 steps for four of six states; Texas and California require more.

We therefore recommend 10,000 steps as the minimum for states with $k \leq 20$. For larger delegations, Table 4 is authoritative for the six studied states: Texas requires 25,000 steps for PP certification, while California requires 50,000. For unstudied large states, a conservative planning rule is

$$n_{\min}(k) = \max(10{,}000, 1{,}000k),$$

followed by actual diagnostic certification. The formula is a pre-run budget heuristic, not a substitute for $\hat{R}$, ESS, and Hamming checks.

7.4. Known Error Rate

Under *Daubert*, an expert method should expose its known or knowable error rate. For an ensemble percentile claim, there are two separate rates. First, the nominal tail threshold gives the false-positive scale: a neutral plan will fall beyond the 95th percentile about 5% of the time under the certified reference distribution, and beyond the 99th percentile about 1% of the time. Second, finite-chain uncertainty depends on ESS, not nominal sample count. A report should therefore state both the percentile threshold and the ESS-based uncertainty interval before using the ensemble as legal evidence.

7.5. ConvergenceSweep is Cheaper and More Transparent

For the specific task of generating the DIA statutory map, the ConvergenceSweep algorithm (U.1) is unambiguously preferable:

- Runtime: $O(T \cdot n)$ with $T = 600$ and n the number of tracts. For Texas: $\approx 600 \times 8{,}000 = 4.8M$ METIS calls. In practice, ConvergenceSweep completes in under 60 seconds for any state.

- Certification: the stopping criterion is a finite inequality check, not an asymptotic convergence argument.

- Auditability: the sequence of seeds, the objective values, and the final map are all deterministic given the census release identifier.

Ensemble sampling requires hours of computation and a convergence certificate based on asymptotic theory. For plan generation, ConvergenceSweep is both cheaper and more transparent.

8. Implementation in `BISECT research`

8.1. Command-Line Interface

The diagnostic math described in this paper is implemented in the `bisect-analysis` crate. The stable command-line surface for metric traces is `BISECT research validate-ensemble`. It consumes one JSONL file per chain, where each line contains a step number and a metric map.

```
BISECT research validate-ensemble \
    --ensemble-dir outputs/ensembles/nc_2020/chains \
    --output validate-ensemble.json \
    --rhat-threshold 1.1 \
    --ess-min 500 \
    --strict
```

The command:

1. Reads every `*.jsonl` file in `-ensemble-dir` as one chain.
2. Validates that R-hat has enough equal-length chains.
3. Computes $\hat{R}$ and ESS for each metric present in every chain.
4. Optionally records enacted-plan percentile ranks when supplied.
5. Reports pass/fail against the explicit thresholds from Section 7.

8.2. Output Format

The output is a JSON file with the following structure:

```
{
  "state": "NC",
  "year": 2020,
  "k": 14,
  "chains": 10,
  "steps": 10000,
  "diagnostics": {
    "rhat_pp": 1.02,
    "rhat_seats": 1.03,
    "ess_pp": 695,
    "ess_seats": 1538,
    "hamming_lag1": 0.87
  },
  "thresholds": {
    "rhat_max": 1.1,
    "ess_min": 500,
    "hamming_max": 0.95
  },
  "verdict": "PASS"
}
```


A `PASS` verdict indicates that the supplied metrics meet the explicit thresholds used for this validation run. It is evidence about convergence of the recorded ensemble metrics, not a standalone legal conclusion about the enacted plan.

8.3. Implementation Details

ReCom proposal. The ReCom step selects two adjacent districts uniformly at random, merges their tract sets, generates a random spanning tree of the merged subgraph, and bisects the spanning tree at its centroid edge subject to population balance. The implementation uses the BISECT spanning-tree and ensemble components when chains are generated inside the workspace.

R-hat computation. Each chain's PP time series is stored in a ring buffer. $\hat{R}$ is computed from the last half of each chain (disarding the burn-in), following the recommendation of Vehtari et al. [2021].

ESS computation. ESS uses the initial monotone sequence estimator of Geyer [1992] to truncate the autocorrelation sum before it becomes noisy.

Hamming autocorrelation. The Hungarian algorithm for label canonicalization is $O(k^3)$ per plan pair. For large k we use the greedy approximation (sort districts by centroid proximity), which is accurate to within 1% for nearly-converged chains.

8.4. Relation to Diagnostics Code

The diagnostics code is exposed through the BISECT analysis crate:

```
bisect_analysis::ensemble_diagnostics::{
    rhat,          // Gelman-Rubin R-hat
    ess,           // Effective sample size
    hamming_lag1,  // Lag-1 Hamming autocorrelation
    diagnose,      // Run all three and return verdict
}
```

The module has 21 unit tests covering boundary cases (all chains identical, all chains diverged, single-chain degenerate input) and is part of the standard `cargo test` suite.

References

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